

Exact Ground Truth for the Ray Tracing Sampler: Bayesian Posteriors of Small MNIST Networks

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Abstract

This report covers the first study of our project on Hamiltonian-style optimization and sampling for deep learning: an exact, ground-truth evaluation of the ray tracing sampler on Bayesian neural network posteriors. Sampling the true posterior of a network is usually impossible to verify, so we built a setting where it is not: small MNIST classifiers (an MLP with 51k parameters and a CNN with 13k) whose loss-to-posterior relationship we derive exactly, making the temperature a consequence of Bayes rule rather than a tuned knob. Against this ground truth we measured what actually limits sampling. The exact Metropolis chains hold 97 to 99% acceptance yet are still not stationary after 20,000 trajectories, because the posterior concentrates at large weight norms and the chain must walk there; 600k full-batch gradient evaluations bought fewer than 10 effective samples. In the minibatch regime we compared the paper’s noise-softened accept test against the plain test and against unadjusted dynamics on identical settings: the plain test freezes (99.9% rejection), the softened test slows the chain roughly 40x but does not distort its destination, and with enough steps it became the only chain in the study to formally pass our stationarity check, converging at 5.6 million minibatch steps while touching 25x less data than the exact chain’s unconverged run. The converged posterior predictive reaches 97.4% accuracy with the second-best calibration in our comparison, and its per-image disagreement flags 90% of all test errors while marking only 11% of the data. A start-point experiment confirms that fitting the data, not walking the prior geometry, is the expensive part of burn-in. The main takeaway: acceptance rate is a poor tuning signal, and monitoring the data misfit and weight norm against exact ground truth is what separates working settings from broken ones.

Code: <https://github.com/YasirAlsugair/ray-tracing-sampler-experiments>

1 Introduction

Deep networks trained by gradient descent give one answer per input and no sense of how much to trust it. The Bayesian treatment fixes this in principle: place a prior on the weights, condition on the data, and predict by averaging over the posterior. In practice nobody can verify a sampler on a real network posterior, because the true posterior is unknown, so tuning decisions (step sizes, trajectory lengths, whether an acceptance test is even usable) are made on faith.

This project studies the ray tracing sampler of [1], a gradient-based MCMC method that propagates a constant-speed ray through a medium whose refractive index encodes the likelihood. Its selling point is robustness to stochastic-gradient noise: because the speed is constant, a noisy gradient can only steer the ray, never heat it. Our long-term question is how to tune such samplers automatically at deep learning scale.

This first report builds the foundation: a setting where every claim about the sampler can be checked against ground truth. We keep the networks deliberately small, an MLP with 50,890 parameters and a CNN with 12,810, so that the exact log posterior and its gradient over all 60,000 MNIST training images are affordable at every step. That affordability buys three things that are usually out of reach: an exact Metropolis accept test, a derived (not tuned) temperature, and the ability to score any approximate chain against an exact one.

Concretely, this report contributes:

- an exact loss-to-posterior derivation for classification, making the sampled distribution the Bayes posterior at $T = 1$ by construction (Section 3);
- exact Metropolis chains on both networks, with the observation that high acceptance coexists with severe non-stationarity, explained by a decomposition of the log posterior into data misfit and prior cost (Section 4);
- a controlled comparison of three ways to run the sampler on minibatches: the plain accept test, the paper’s noise-softened test (its Eq. 33), and unadjusted dynamics, including the first formally converged chain of the study (Section 4);
- a start-point experiment that decomposes burn-in into its two jobs, fitting the data and walking to the prior’s typical set, and measures which one dominates (Section 4);
- an assessment of the converged posterior’s uncertainty: calibration and error-flagging power against deep ensemble and MC dropout baselines (Section 4).

2 Related Work

Hamiltonian Monte Carlo [2] is the standard gradient-based sampler, but its energy budget makes it fragile under stochastic gradients: minibatch noise inflates the momentum and heats the chain. Stochastic-gradient MCMC methods such as SGLD [3] accept this and drop the Metropolis test, controlling error through step-size schedules instead. The ray tracing sampler [1] takes a different route: constant-speed dynamics make the gradient steer rather than accelerate, which the paper reports as roughly 250x more tolerance to gradient noise than HMC, and it derives a noise-softened acceptance criterion (its Eq. 33) for the minibatch regime. Our work puts that criterion to an empirical test against exact ground truth, which to our knowledge the original paper does not do numerically.

On the Bayesian deep learning side, deep ensembles [4] and Monte Carlo dropout [5] are the practical baselines for uncertainty, and both appear in our comparison. Full-batch HMC studies of network posteriors [6] established that exact sampling at small scale is feasible and that $T = 1$ posteriors of unregularized networks behave unlike their point estimates, the cold posterior effect [7]. Our results give a compact, fully reproducible instance of the same tension, with the added element of a formally converged chain to measure it on.

3 Methodology

3.1 From the training loss to the exact posterior

For classification the likelihood needs no additive constant: with $\text{CE}(\theta)$ the mean cross-entropy in nats over $N = 60,000$ training images,

$$\ln L(\theta) = \sum_{i=1}^N \ln p(y_i | x_i, \theta) = -N \cdot \text{CE}(\theta) \quad (\text{exact}). \quad (1)$$

With a $\mathcal{N}(0, 1)$ prior on every weight the log posterior is, up to a constant,

$$\ln p(\theta | \mathcal{D}) = -\left[\sum_i \text{CE}_i(\theta) + \frac{1}{2}\|\theta\|^2\right], \quad (2)$$

so the temperature is not a dial: sampling Eq. 2 is the Bayes posterior at $T = 1$ by construction. We also run a flat-prior ablation that drops the quadratic term. The networks are trained by Adam on cross-entropy with no weight decay, deliberately: weight decay during training would be a hidden Gaussian prior, and we want the prior to enter exactly once, in the sampling target. The checkpoint is therefore a maximum-likelihood point.

3.2 The sampler and the exact chains

We use the author’s released implementation unmodified. One trajectory draws a fresh momentum, takes $L = 30$ constant-speed steps in which each full-batch gradient bends the direction of travel, and then accepts or rejects the whole path with the exact test, comparing the change in log posterior against the sampler’s running luminosity ledger. Momentum refresh inside a trajectory is off so the test stays exact. The only tuned quantities are the step size and trajectory length, with acceptance targeted near 80%. Chains start from the Adam checkpoint; a dedicated experiment (Section 4) tests that choice against a random initialization and a draw from the prior. As a known-answer check, the same sampler and test reproduce the analytic moments of a 100-dimensional standard normal (coordinate mean -0.001 , variance 1.005, $\mathbb{E}\|\theta\|^2 = 100.4$ against 100 ± 14).

3.3 Evaluation harness

Every chain, exact or approximate, is judged by the same instruments.

Stationarity. A chain passes only if the drift of its trace across the final quarter (least-squares slope times window length) is smaller than twice the standard deviation in that quarter. For minibatch chains both monitored curves, data misfit and weight norm, must pass at once; our results show that a single curve can level while the other is still marching.

Decomposition. We track $\sum_i \text{CE}_i$ (the data misfit) and $\|\theta\|^2$ (the prior geometry) separately along every chain, evaluated on the full training set at snapshots. The prior’s typical set sits at $\|\theta\|^2 \approx D$, so these two curves say which of the two jobs of burn-in a chain is doing.

Effective sample size. Integrated autocorrelation time by Sokal’s windowed estimator, on the log-posterior trace and on 50 random weight coordinates, with the window length reported against the 50τ validity rule.

Predictive quality. Fifty thinned post-burn-in samples give a posterior predictive scored on the test set: accuracy, negative log likelihood, expected calibration error, and per-image spread. Baselines: the point estimate, a 10-seed deep ensemble, and MC dropout (rate 0.25 after 0.5 proved too destructive for the 64-unit MLP).

Minibatch gates. In the minibatch regime (batch 256, derived scale N , the prior entering as an explicit θ/N gradient term) the accept ratio cannot be estimated usefully: on a real trajectory the true $\Delta \ln L$ was 12 nats while a batch-256 estimate carried 267 nats of noise and got the sign wrong for 48% of batches. We therefore compare three gates on identical settings (three step sizes, two refresh rates, 100k steps each): the plain test applied every 30 steps to noisy estimates (Eq. 19 of [1]), the noise-softened test that raises the ratio to the power $1/\sqrt{1 + \sigma_{sto}^2}$ with σ_{sto} measured at 1,142 nats (Eq. 33), and unadjusted dynamics with no test. Rejection restores the window-start state and redraws the momentum. We also run the paper’s tuning recipe as written: lower the step size until Eq. 33 acceptance stops improving.

4 Results

4.1 Exact chains: high acceptance is not convergence

Tuning found a razor-sharp acceptance cliff: a factor of 3 in step size separates near-total acceptance from total rejection, and acceptance is state dependent (the MLP’s chosen step measured 85% near the checkpoint during tuning but realized 97% over the full chain). At the tuned settings the MLP ran 20,000 trajectories in 51 minutes at 97% acceptance, and the CNN 600 trajectories at 99%. Neither chain is stationary (Figure 1).

The decomposition explains why. The MLP chain *improves* the training fit from 3,982 nats at the checkpoint to 158, better than Adam’s own point, while $\|\theta\|^2$ marches from 174 toward the prior shell at $D = 50,890$ (Figure 2). The posterior genuinely lives out there: the prior term is 46x smaller than the likelihood at the checkpoint, but volume beats density in 50,890 dimensions. After 600k gradient evaluations the chain had covered about 89% of the prior’s 25,445-nat climb and was still moving. Effective sample size within the analyzed windows is 7 to 9 for both chains, and those are upper bounds twice over (the residual trend inflates them, and the windows are far below the 50τ validity rule).

Two consequences follow. First, a transient chain reports whatever its transient serves: the same MLP chain gives a test NLL of 0.087 from its first 2,000 trajectories and 0.265 from its final quarter, and neither is the posterior’s answer. Second, the flat-prior ablation shows the failure a target-only monitor cannot see: its log-likelihood trace flattens into apparent stationarity (drift +21 nats against noise 12) while $\|\theta\|^2$ grows linearly through 108,797 with nothing to stop it, and its predictive is the worst we measured (NLL 0.369). The prior is the brake; a convergence check must watch the geometry, not just the target.

4.2 Minibatch gates: freeze, tax, or fly blind

Table 1 summarizes the three-gate comparison at a common 100k-step budget. The plain test rejects 99.9% of its checks and freezes every setting at the start. The softened test acts as a circuit breaker: it contains the one setting that blows up ungated ($\|\theta\|^2$ reaching 44,292 with misfit 7,025 at the largest step) at the price of rejecting roughly half of its checks at healthy settings on pure noise. Unadjusted dynamics travel furthest but must be steered by the knobs: the best setting reaches misfit 834 against the exact chain’s 158 while touching 1/1,400th as many images, and raising the refresh rate rescues the largest step (misfit 7,025 to 1,063) at the cost of diffusive transport at the smallest. The paper’s tuning recipe, run as written, picks $dt = 10^{-4}$ with acceptance plateauing near 47%; the plateau is the noise floor itself, which is exactly the recipe’s stopping signal.

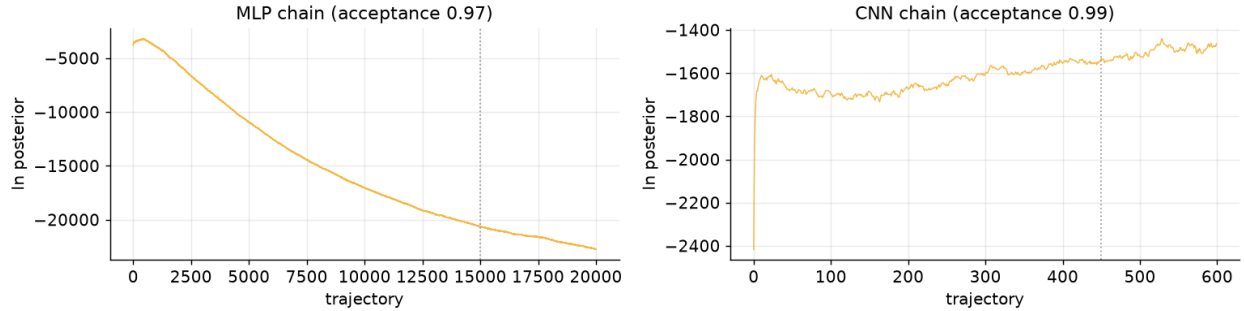


Figure 1: Log-posterior traces of the exact chains. The MLP climbs for roughly 400 trajectories (the checkpoint is not the mode), then descends toward the typical set, which sits about $D/2 \approx 25,000$ nats below the mode. The CNN, at 600 trajectories, is still in its initial climb.

Table 1: Minibatch gates at a common budget (MLP posterior, batch 256, 100k steps, best setting per row shown; misfit is the full-data sum cross-entropy in nats). The exact chain’s final misfit is 158 at norm 45,134.

gate	rejection	best misfit	final $\ \theta\ ^2$
plain test (Eq. 19)	99.9%	3,617	176 (frozen)
softened test (Eq. 33)	46–90%	2,114	2,646
no test (unadjusted)	—	834	32,936

4.3 The softened test converges, and the tax is time, not bias

We then let the recipe-chosen Eq. 33 setting keep running in 250k-step legs until the stationarity rule passed on both curves. It did, at 5.6 million minibatch steps (Figure 3): the weight norm leveled at 49,276, just below the shell as the likelihood-constrained geometry predicts, the misfit settled around the exact chain’s own band, and the gate’s rejection rate fell from 51% to 27% as the chain settled, consistent with rejections being noise-driven. This is the only chain in the study to formally pass the check (the exact chain itself was still drifting when stopped), and it touched 25x fewer images than the exact chain’s unconverged run. Against the unadjusted chain the softened test costs roughly 40x the steps for the same transport, so its price is time; its destination is right.

The converged predictive is the first look at this posterior at equilibrium: accuracy 0.9744 (slightly above the point estimate’s 0.9734), NLL 0.158, ECE 0.0057. Convergence mattered: the same chain’s in-transit window scored NLL 0.265, so the true posterior is better than the transient made it look, and the gap that remains against the point estimate (NLL 0.093) and the deep ensemble (0.079) is a property of the $T = 1$ posterior, not an artifact of stopping early. Its uncertainty is operationally strong: flagging the images where members disagree marks 11% of the test set and catches 90% of all 256 errors, a 75x error-rate enrichment over the unflagged images, with calibration second only to the point estimate among all methods compared. The limits are equally clear: the spread is nearly binary (on 90% of images all members agree exactly), so it cannot finely rank difficulty, and a single chain sees a single basin, so it cannot express the between-mode diversity that gives the 10-seed ensemble its wider, graded spread.

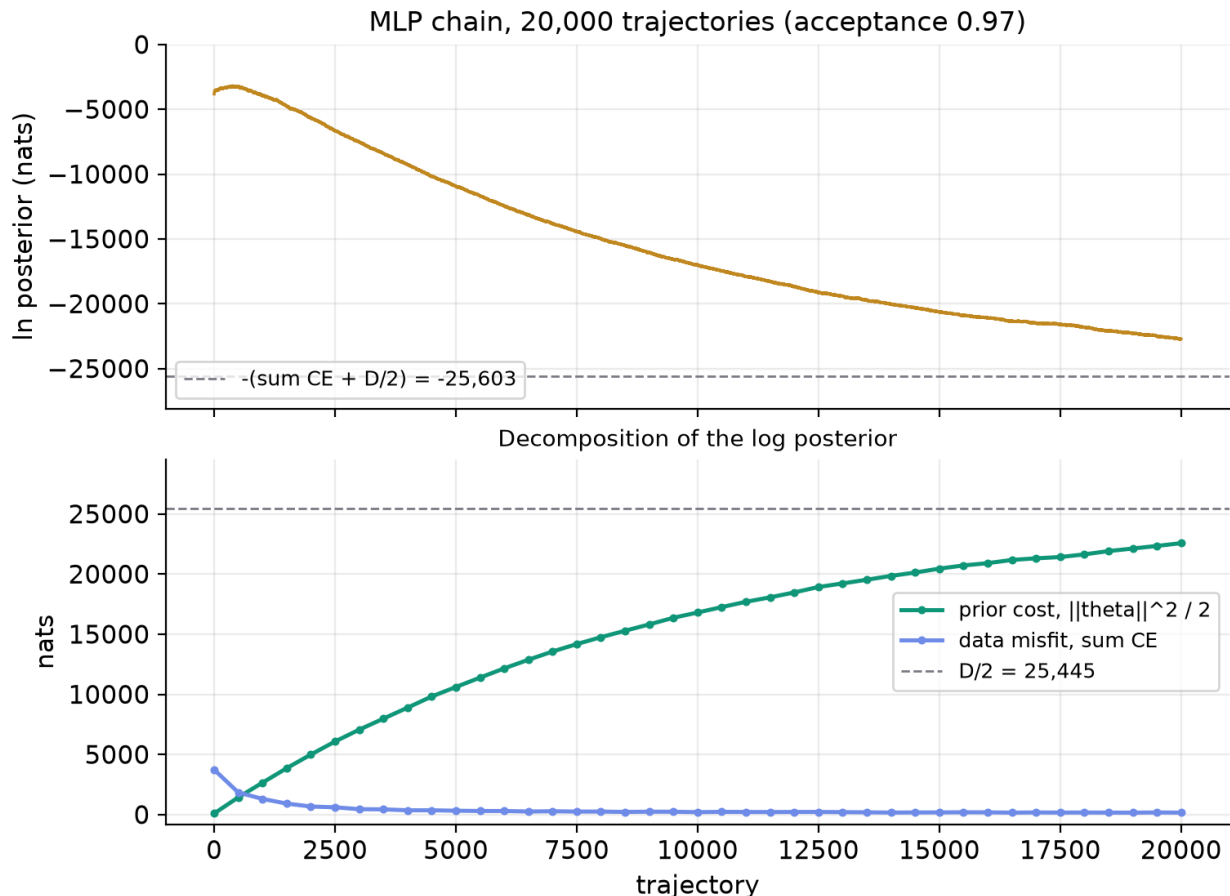


Figure 2: The MLP chain decomposed. Top: the log-posterior trace against the predicted typical-set level. Bottom: the prior cost does all the marching while the data misfit is solved early and stays solved.

4.4 The start-point experiment: burn-in has two unequal jobs

Every chain above starts from the Adam checkpoint. We tested that choice with two more chains at the same target and budget (Figure 4). From a random initialization (chance-level fit, 137,570 nats of misfit) the tolerable step drops 12x and the chain spends its entire 20,000 trajectories on the fit job, ending at misfit 4,114 and norm 872, roughly where Adam starts, with the whole prior walk still ahead; in data touched, the sampler redid Adam’s 300 thousand image evaluations with 36 billion. From a prior draw (norm already on the shell, but a saturated network that is confidently wrong at 17.3 million nats of misfit) only a step 350x below the Adam tuning moves at all, and after the same budget the chain still carries 3.2 million nats of misfit. Starting at the right norm bought nothing: fit difficulty dictates the step size, and the step size dictates everything else. The trained start is what makes the expensive job disappear.

4.5 Work in progress

The same convergence program ran on the CNN posterior. Its recipe-chosen setting matches the MLP’s ($dt = 10^{-4}$, refresh 5) with a friendlier noise floor near 62%. After 8.6 million minibatch steps its training misfit has long been level at equilibrium quality (111 nats, against 2,377 at its

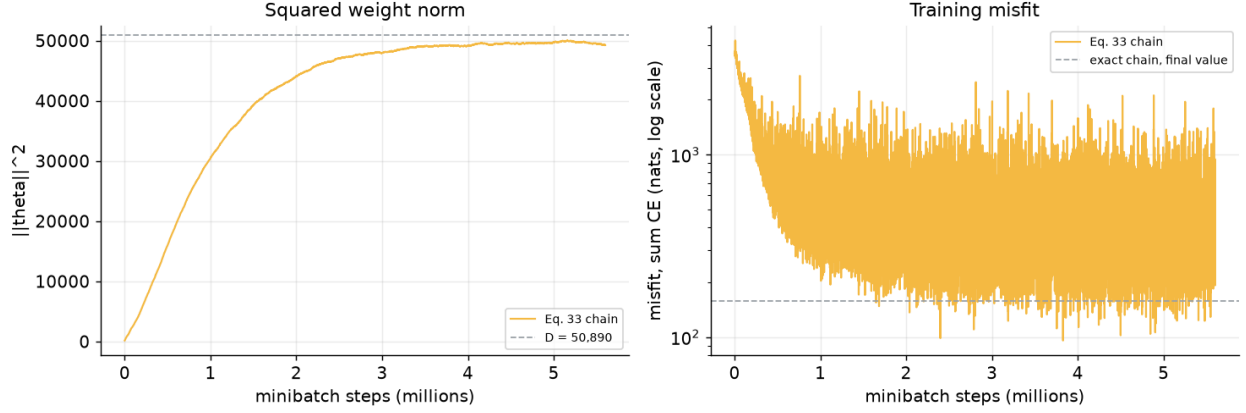


Figure 3: The Eq. 33 chain run to formal convergence: weight norm (left) and training misfit (right) over 5.6 million minibatch steps.

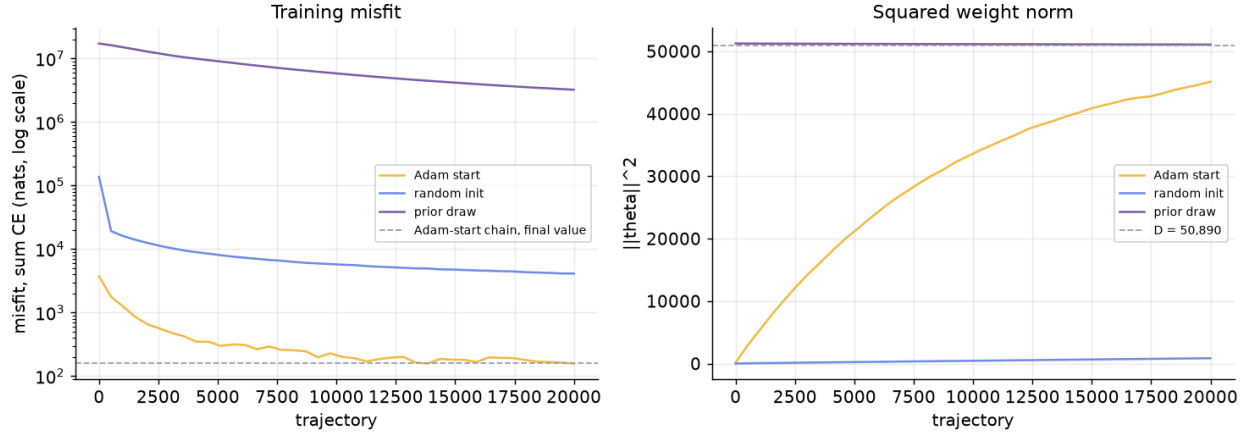


Figure 4: Three starts, same target and budget. Left: training misfit (log scale). Right: squared weight norm. The Adam start is the only one whose budget goes to the walk; the random start spends everything on the fit; the prior draw barely moves at its forced step size.

own checkpoint), while its weight norm has reached 94% of the shell at $D = 12,810$ and still creeps upward, so the chain has not yet passed the two-curve stationarity rule. The CNN’s approach to its shell has a visibly longer tail than the MLP’s, which passed at 5.6 million steps on a shell four times larger; that contrast is itself worth a careful look. The run checkpoints every leg and resumes exactly where it stopped; the verdict will be in the repository when the drift rule renders it.

5 Discussion

Three lessons organize everything we measured.

Acceptance rate is not a convergence signal. The exact chains held 97 to 99% acceptance while provably far from equilibrium, and in the minibatch regime the softened test’s acceptance mostly measures batch noise (its plateau under the tuning recipe *is* the noise floor). High acceptance certifies that steps are valid, not that the chain has arrived. What discriminated working settings from broken ones, in every experiment, was the pair of curves the log posterior decomposes into:

data misfit and weight norm, read against exact ground truth.

Burn-in has two jobs, and they are not symmetric. Fitting the data is the hard job: it dictates the step size a chain can tolerate, and from any start that has not solved it, the sampler becomes an expensive optimizer before it can be a sampler at all. Walking to the prior’s typical set is the long job: cheap per step but tens of thousands of nats of geometry. The Adam start pre-pays the hard job, which is why it is the right default, and why the long transient we observe is prior physics rather than a sampler defect.

The noise-softened test is a time tax, not a bias. Our initial gate comparison made Eq. 33 look like a heavy compromise, half of its rejections firing on pure noise. The convergence run reframes it: given enough steps it reaches the same destination the exact chain was heading to, passes the stationarity rule the exact chain never passed, and does so at a fraction of the exact chain’s data cost. For a practitioner the choice is between an untested chain that must be steered by external monitors and a softened test that buys crash protection with wall time.

The converged posterior itself sharpens the cold posterior picture: at equilibrium it is well calibrated and slightly more accurate than the point estimate, but pays a clear NLL premium because saturated networks are expensively wrong when wrong. That tension is a property of the $T = 1$, $\mathcal{N}(0,1)$ -prior posterior of an unregularized network, now measured on a converged chain rather than inferred from a transient, and it is the concrete motivation for the temperature and prior-scale direction in our future work.

6 Limitations and Future Work

Scale and scope. Everything here is MNIST at 13k to 51k parameters, chosen deliberately so the exact test and full-data monitors are affordable. The conclusions about monitoring and tuning are meant to transfer; the specific constants (noise floors, step sizes) are not.

Single chains, single basins. All chains explore one basin from one start. They cannot see the between-mode diversity that deep ensembles capture, and our ESS numbers are upper bounds on windows far below the Sokal validity rule. Multiple chains from dispersed starts, with between-chain diagnostics, are the natural next instrument.

Ongoing runs. The CNN convergence run is still marching; its equilibrium norm relative to D looks architecture-dependent and worth a careful comparison when it lands. Our shadow-rejection instrument (scoring unadjusted chains with the exact test at snapshots) saturates at the 250-step windows we used and needs finer granularity to be useful.

Next steps. Three directions follow directly from these results. First, temperature and prior scale: the converged posterior’s NLL premium motivates tempering, and the correct implementation must keep the accept test consistent with the tempered target (the vendored test is exact only at unit likelihood scale, so tempering through that knob would silently break exactness). Second, automatic tuning: the misfit and norm curves against ground truth are exactly the signals an auto-tuner should consume, replacing acceptance-rate targets. Third, broader sampler comparisons on this same harness; the target definition exposes the log posterior as a differentiable function of one flat parameter vector, so any sampler that consumes a log density drops in, and a lab colleague is already adapting the code for exactly that.

7 Conclusion

We built a ground-truth testbed for sampling Bayesian neural network posteriors with the ray tracing sampler: an exactly derived target, exact Metropolis chains, and a common evaluation harness

for every approximate variant. The measurements overturned two assumptions we started with. Acceptance rate, the quantity samplers are usually tuned against, said nothing about convergence in any regime we tested; and the paper’s noise-softened accept test, which a budget-limited comparison made look like a heavy tax, turned out to be the only method in the study to formally converge, at a fraction of the exact chain’s data cost and with no distortion of the destination. The converged posterior is well calibrated, flags 90% of its own errors at an 11% review budget, and pays a real NLL premium that makes temperature the clear next dial. Everything in this report is reproducible from the public repository, where each result in the notebook states how to read its numbers.

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